

Marcelo Bongarti, Ph.D.

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Employment History

- 06/2021 – present 📌 **Postdoctoral Researcher**, Weierstrass Institute for Applied Analysis and Stochastics, Berlin, Germany.
Projects: “Berlin Cluster of Excellence Math+ (Project AA4-3)”
“SFB-TRR 154 (Projects B02 and C08)”
- 05/2020 – 08/2020 📌 **NSF-MSGI Summer Research Internship**, University of California, Los Angeles, LA, USA. Mentor: *Andrea L. Bertozzi*.
- 05/2019 – 08/2019 📌 **NSF-MSGI Summer Research Internship**, Argonne National Laboratory, Chicago, IL, USA. Mentors: *Paul Hovland and Sri H. K. Narayanan*.

Education

- 2016 – 2021 📌 **Ph.D., Mathematics**
Department of Mathematical Sciences - University of Memphis, GPA 3.99
Thesis title: *On nonlinear acoustic waves of third-order (in time) and applications*.
Advisors: *Irena Lasiecka & Roberto Triggiani*
- 2014 – 2016 📌 **M.Sc. Mathematics**
Department of Mathematics - Institute of Biosciences, Languages and Exact Sciences of the São Paulo State University, GPA 3.88
Thesis title: *Domains of Fractional Powers of Operators*.
Advisor: *German Lozada-Cruz*
- 2010 – 2014 📌 **B.Sc. Mathematics**
Department of Mathematics - Institute of Biosciences, Languages and Exact Sciences of the São Paulo State University

Research Interests

- ▶ Parabolic and hyperbolic PDEs. ▶ PDE constrained optimization.
▶ Boundary control and stabilization of PDEs. ▶ (Generalized) Nash equilibrium problems.

Awards and Achievements

- 2021 📌 **Morton Thesis Award (nominated)**, University of Memphis.
- 2020 📌 **Dr. Ralph Faudree Research Award**, University of Memphis.
📌 **NSF-MSGI Summer Research Internship**, University of California, Los Angeles.
- 2019 📌 **Student Research Forum Award for Best Poster**, University of Memphis.
📌 **NSF-MSGI Summer Research Internship**, Argonne National Laboratory.
- 2014 📌 **National Journey of Scientific Initiation**, National Institute for Pure and Applied Mathematics (IMPA).
- 2006-2009 📌 **Brazilian Olympiad of Mathematics for Public Schools (Honor Mention)**

Organizational Activities

- ▶ Organizer of the mini symposium *Modeling, Analysis and Optimal Control of Infinite Dimensional Problems and Applications* at the IFIP TC7 conference in Hamburg, in August of 2024.
- ▶ Local organizer of the 15th International Conference on Free Boundary Problems: Theory and Applications 2021 (FBP 2021), virtual conference between 13-17th of September, 2021.
- ▶ Organizer of the IV Mini-workshop on PDEs of the Department of Mathematics of the Institute of Biosciences, Languages and Exact Sciences – São Paulo State University (Brazil), on March 18th, 2021.

Research Publications

Preprints

- 1 A. Alphonse, **M. Bongarti**, P. Jaap, C. N. Rautenberg, and E. Valdinoci, *A nonlocal distance-based alt-caffarelli*, Preprint in preparation, 2026.
- 2 **M. Bongarti** and M. Hintermüller, *State constrained convex nash equilibrium problems coupled with linear hyperbolic PDEs*, Submitted to *Siopt*, 2026.
- 3 **M. Bongarti** and M. Hintermüller, *Structure versus regularity of set-valued maps in convex generalized nash equilibrium problems in banach spaces*, Submitted to *Systems & Control Letters*, 2026.
- 4 **M. Bongarti**, F. Meyer, and J. Müller, *Reinforcement learning for network cybersecurity with asymmetric attacker & defender dynamics*, In preparation, 2026.
- 5 **M. Bongarti** and I. Lasiecka, *The nonlinear JMGT equation with dynamic boundary conditions and degenerate viscoelasticity: Asymptotic analysis w.r.t time relaxation*, Preprint in preparation, 2026.
- 6 **M. Bongarti**, I. Lasiecka, and S. Silva, *Asymptotic analysis (w.r.t thermal relaxation) for MGT equation with partially dissipative boundary data and degenerate viscoelasticity*, Preprint in preparation, 2026.

Journal Articles

- 1 **M. Bongarti**, C. Parkinson, and W. Wang, “Optimal control of a reaction-diffusion epidemic model with non-compliance,” *European Journal of Applied Mathematics*, pp. 1–26, 2025. [DOI](#): 10.1017/S0956792525000130.
- 2 R. C. Contreras, M. S. Viana, E. S. Fonseca, **M. Bongarti**, O. Toygar, and R. C. Guido, “Exploring multicepstral features in a new classical machine learning-based framework for replay attack detection,” *Computers & Electrical Engineering*, 2025, Accepted, to appear.
- 3 **M. Bongarti** and M. Hintermüller, “Optimal boundary control of the isothermal semilinear Euler equation for gas dynamics on a network,” *Applied Mathematics and Optimization*, vol. 89, no. 36, 2023. [DOI](#): 10.1007/s00245-023-10088-0.
- 4 A. Bertozzi, **M. Bongarti**, L. D. Galvan, L. Hatcher, M. R. Lindstrom, C. Parkinson, and C. Wang, “Alternative SIAR models for infectious diseases and applications in the study of non-compliance,” *Mathematical Models and Methods in Applied Sciences*, vol. 32, no. 10, pp. 1987–2015, 2022. [DOI](#): <https://doi.org/10.1142/S0218202522500464>.
- 5 **M. Bongarti** and I. Lasiecka, “Boundary feedback stabilization of a critical nonlinear JMGT equation with Neumann-undissipated part of the boundary,” *Discrete and Continuous Dynamical Systems - S*, vol. 15, no. 8, pp. 1957–1985, 2022. [DOI](#): 10.3934/dcdss.2022107.
- 6 **M. Bongarti**, I. Lasiecka, and J. H. Rodrigues, “Boundary stabilization of the linear MGT equation with partially absorbing boundary data and degenerate viscoelasticity,” *Discrete and Continuous Dynamical Systems - S*, vol. 15, no. 6, pp. 1355–1376, 2022. [DOI](#): 10.3934/dcdss.2022020.
- 7 **M. Bongarti**, I. Lasiecka, and R. Triggiani, “The SMGT equation from the boundary: regularity and stabilization,” *Applicable Analysis*, vol. 101, no. 5, pp. 1735–1773, 2021. [DOI](#): 10.1080/00036811.2021.1999420.
- 8 **M. Bongarti**, S. Charoenphon, and I. Lasiecka, “Vanishing relaxation time dynamics of the Jordan Moore-Gibson-Thompson equation arising in nonlinear acoustics,” *Journal of Evolution Equations*, 2020. [DOI](#): 10.1007/s00028-020-00654-2.

Conference Proceedings

- 1 R. C. Contreras, G. L. Heck, M. S. Viana, **M. Bongarti**, H. Zamani, and R. C. Guido, “Metaheuristic Algorithms for Enhancing Multicepstral Representation in Voice Spoofing Detection: An Experimental Approach,” in *Advances in Swarm Intelligence*, Y. Tan and Y. Shi, Eds., Singapore: Springer Nature Singapore, 2024, pp. 247–262, ISBN: 978-981-9771-81-3.

- 2 R. C. Contreras, M. S. Viana, P. C. Pessoa, **M. Bongarti**, H. Zamani, and R. C. Guido, “Enhancing metaheuristic algorithms for voice spoofing detection via multi-cepstral representation,” in *Advances in Swarm Intelligence: 15th International Conference, ICSI 2024, Shenzhen, China, July 15–18, 2024, Proceedings, Part I*, Y. Tan and Y. Shi, Eds., ser. Lecture Notes in Computer Science, vol. 14822, Cham: Springer Nature Switzerland, 2024, pp. 247–262, ISBN: 978-3-031-84355-6. [DOI: 10.1007/978-3-031-84356-3_19](#).
- 3 M. S. Viana, R. C. Contreras, P. C. Pessoa, **M. Bongarti**, H. Zamani, R. C. Guido, and O. Morandin Junior, “Massive Conscious Neighborhood-Based Crow Search Algorithm for the Pseudo-Coloring Problem,” in *Advances in Swarm Intelligence*, Y. Tan and Y. Shi, Eds., Singapore: Springer Nature Singapore, 2024, pp. 182–196, ISBN: 978-981-9771-81-3.
- 4 S. H. K. Narayanan, T. Propson, **M. Bongarti**, J. Hüchelheim, and P. Hovland, “Reducing memory requirements of quantum optimal control,” in *Computational Science – ICCS 2022*, D. Groen, C. de Mulatier, M. Paszynski, V. V. Krzhizhanovskaya, J. J. Dongarra, and P. M. A. Sloot, Eds., Springer, 2022, pp. 129–142, ISBN: 978-3-031-08760-8. [DOI: 10.1007/978-3-031-08760-8_11](#).
- 5 **M. Bongarti**, I. Lasiecka, and J. H. Rodrigues, “Optimal boundary feedback control of high intensity focused ultrasound processes,” in *60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 3318–3323. [DOI: 10.1109/CDC45484.2021.9683592](#).

Books and Chapters

- 1 **M. Bongarti** and I. Lasiecka, “Boundary Stabilization of the linear MGT equation with Feedback Neumann control,” in *Deterministic and Stochastic Optimal Control and Inverse Problems*, B. Jadamba, A. A. Khan, S. Migórski, and M. Sama, Eds., CRC Press, 2021, ch. 7, pp. 150–168. [URL: 10.1201/9781003050575](#).
- 2 **M. Bongarti**, S. Charoenphon, and I. Lasiecka, “Singular Thermal Relaxation Limit for the Moore-Gibson-Thompson Equation Arising in Propagation of Acoustic Waves,” in *Semigroups of Operators – Theory and Applications*, J. Banasiak, A. Bobrowski, M. Lachowicz, and Y. Tomilov, Eds., vol. 325, Springer, 2020, pp. 147–182. [DOI: 10.1007/978-3-030-46079-2_9](#).

Teaching & mentoring experience

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| 2023 | <p>■ Weierstrass Institute for Applied Analysis and Stochastics
 <i>Berlin, Germany</i></p> <p>▶ Mentor (alongside Prof. Michael Hintermüller) of Ms. Evaëlle Riffard at the “WIAS internship program” (Summer)</p> |
| 2021 | <p>■ Department of Mathematical Sciences
 <i>University of Memphis, Memphis, TN, USA</i></p> <p>▶ Mentor at the “NSF–STEM collaborative research initiative” (Spring)</p> |
| 2020 | <p>■ Department of Mathematical Sciences
 <i>University of Memphis, Memphis, TN, USA</i></p> <p>▶ Independent instructor for “Differential and Integral Calculus I” (Spring)</p> <p>▶ Responsible for the media preparation for “Differential and Integral Calculus III” (Spring)</p> <p>▶ Independent instructor for “Differential and Integral Calculus I” (Fall)</p> <p>▶ Instructor (joint work with Prof. Irena Lasiecka) for “Distribution Theory and Sobolev Spaces” (Fall)</p> <p>▶ Mentor at the “NSF–STEM collaborative research initiative” (Spring and Fall)</p> |
| 2019 | <p>■ Department of Mathematical Sciences
 <i>University of Memphis, Memphis, TN, USA</i></p> <p>▶ Independent instructor for “Ordinary Differential Equations” (Spring)</p> <p>▶ Mentor at the “NSF–STEM collaborative research initiative” (Summer)</p> <p>▶ Independent instructor for “Differential and Integral Calculus I” (Fall)</p> |
| 2018 | <p>■ Department of Mathematical Sciences
 <i>University of Memphis, Memphis, TN, USA</i></p> <p>▶ Independent instructor for “Elementary Calculus” (Fall)</p> <p>▶ Independent instructor for “Elementary Calculus” (Summer)</p> <p>▶ Responsible for the “Analysis PhD Qualifying Exam” Review Sessions (Summer)</p> |

Teaching & mentoring experience (continued)

- 2017  **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
Independent instructor for “Trigonometry” (Fall)
- 2015  **Institute of Biosciences, Languages and Exact Sciences**
São Paulo State University, S.J. Rio Preto, SP, Brazil
Teaching assistant for “Differential and Integral Calculus I” (Fall)

Selected talks at conferences and seminars

- May, 2026  **MOG 2026 - Mathematics of Networks with Physical Transport**
Novotel Centre Ville, Nürnberg, Germany
Poster presentation
- March, 2025  **WIAS Nonsmooth variational problems and operator equations seminar**
Weierstrass Institute, Berlin, Germany
(Invited) talk: “Multi-objective optimization with linear hyperbolic PDE Constraints: generalized Nash equilibrium problems and gas market applications”
- July, 2023  **SIAM Conference on Control and Its Applications (CT23)**
Sonesta Philadelphia Rittenhouse Square Hotel, Philadelphia, PA, USA
(Invited) talk: “Network Boundary Control of the Semilinear Isothermal Euler Equation Modeling Gas Transport on a Network”
- June, 2023  **93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM 23)**
Technische Universität Dresden, Germany
(Invited) talk: “Network Boundary Control of the Semilinear Isothermal Euler equation Modeling Gas Transport on a Network”
- November, 2022  **Oberseminars Partielle Differentialgleichungen**
Fachbereich Mathematik und Statistik, Universität Konstanz, Konstanz, DE
(Invited) talk: “Boundary stabilization of nonlinear dynamics of acoustics waves under the JMGT equation”
- October, 2022  **Early Career Math Colloquium**
University of Arizona, Tucson, AZ, USA
(Invited, online) talk: “Boundary stabilization of nonlinear dynamics of acoustics waves under the JMGT equation”
- July, 2022  **15th International Conference on Mathematical and Numerical Aspects of Wave Propagation (Waves 22)**
École Nationale Supérieure de Techniques Avancées (ENSTA), Paris, France
(Invited) talk: “Boundary feedback stabilization of a critical nonlinear JMGT equation with Neumann-undissipated part of the boundary”
-  **International Federation of Information Processing, System Modeling and Optimization (IFIP-TC7)**
Warsaw University of Technology, Warsaw, Poland
(Invited) talk: “Boundary feedback stabilization of a critical nonlinear JMGT equation with Neumann-undissipated part of the boundary”
-  **International Federation of Information Processing, System Modeling and Optimization (IFIP-TC7)**
Warsaw University of Technology, Warsaw, Poland
(Invited) talk: “Nonlinear gas transport on a network of pipelines: analysis and control”
- October, 2020  **Web Mathematics Week**
Institute of Biosciences, Languages and Exact Sciences of the São Paulo State University, S.J. Rio Preto, SP, Brazil
(Invited) Mini-course: “Basic compartmental models for Epidemiology”

Selected talks at conferences and seminars (continued)

	■ Dr. Ralph Faudree Lectures <i>University of Memphis, Memphis, TN, USA</i> (Invited) talk: “The Singular Thermal Relaxation Limit for the JMGT-Equation”
September, 2020	■ XIII Mathematics Week <i>Santa Cruz State University, Ilhéus, BA, Brazil</i> (Invited, online) talk: “Basic compartmental models for Epidemiology”
February, 2020	■ Summer Meeting on Differential Equations – 2020 Chapter <i>Institute of Mathematical and Computer Science, São Carlos, SP, Brazil</i> Talk: “Boundary Stabilization of the MGT Equation arising in Nonlinear Acoustics with a Second Sound”
January, 2020	■ III Miniworkshop on PDE’s <i>Institute of Biosciences, Languages and Exact Sciences of the São Paulo State University, S.J. Rio Preto, SP, Brazil</i> (Invited) talk: “Boundary Stabilization of the MGT Equation arising in Nonlinear Acoustics with a Second Sound”
November, 2019	■ PDE Seminar <i>University of Memphis, Memphis, TN, USA</i> (Invited) talk: “Efficient Quantum Optimal Control Algorithms”
March, 2019	■ Workshop on Mathematical Models for Pattern Formation <i>Center for Nonlinear Analysis of the Carnegie Mellon University, Pittsburgh, PA, USA</i> Poster: “The Singular Thermal Relaxation limit of MGT-Equation arising in Non-linear Acoustics”
December, 2018	■ Workshop “Dynamics, Control and Numerics for Fractional PDEs” <i>University of Puerto Rico, San Juan, Puerto Rico</i>
March, 2018	■ Workshop on Mathematical Aspects of Fluid Dynamics (Shanks) <i>Vanderbilt University, Nashville, TN, USA</i>
October, 2017	■ PDE Seminar <i>University of Memphis, Memphis, TN, USA</i> (Invited) talk: “Domain of Fractional Power of Second Order Elliptic Operators with Neumann Boundary Conditions”
April, 2017	■ PDE Seminar <i>University of Memphis, Memphis, TN, USA</i> (Invited) talk: “Fractional Powers of Closed Operators”

Skills

Languages	■ Strong reading, writing and speaking competencies for English (fluent) and Brazilian Portuguese (native).
Coding	■ Python (basic) and L ^A T _E X

Miscellaneous Experience

Reviewer at the following journals

- **SIMA:** SIAM Journal on Mathematical Analysis.
- **ESAIM:** Control, Optimisation and Calculus of Variations.
- **AM:** Acta Mechanica.
- **JOTA:** Journal of Optimization Theory and Applications.
- **DCDS:** Discrete and Continuous Dynamical Systems.
- **AMO:** Applied Mathematics and Optimization.

Certification

Ongoing	■ IBM Data Science with Python , coursera.
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Miscellaneous Experience (continued)

- 📖 **Machine Learning**, coursera.
- 📖 **SQL**, CodeInfinity.

References

- ▶ **Prof. Irena Lasiecka**, University of Memphis
✉ lasiecka@memphis.edu
- ▶ **Prof. Roberto Triggiani**, University of Memphis
✉ rtrggani@memphis.edu
- ▶ **Prof. Andrea Bertozzi**, University of California, Los Angeles
✉ plizardo@math.ucla.edu
- ▶ **Prof. German Lozada-Cruz**, São Paulo State University
✉ german.lozada@unesp.br